QUESTION ANALYSIS

EEE339

Power Electronics

Question Analysis

Sessions 2020–21 *1 papers, 31 questions, 59 formulas*

INSIDE

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1. At a glance:

This analysis covers **1 paper** (2020-21), with 31 questions and 190 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

With one paper, no trend can be read. Use this as a map of the topics.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Power devices	48	25 %	8	Diodes (16); Thyristors (12); TRIAC and BJT (8)
Rectifiers	42	22 %	6	Three-phase (19); Full-wave (13); Half-wave (10)
DC-DC converters	34	18 %	6	Buck (15.5); Boost (11); SEPIC and Cuk (5)
Inverters	27	14 %	4	PWM (10); Filters (7); Multilevel (6)
Power computations	19	10 %	3	Distortion (13); Nonlinear loads (6)

A formula name in bold is a key formula.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	Term tests	All
Power computations	**13**	**6**	**19**
Distortion	**13**	·	13
Nonlinear loads	·	**6**	6
Power devices	**39**	**9**	**48**
Diodes	**12**	**4**	16
Thyristors	**7**	**5**	12
TRIAC and BJT	**8**	·	8
IGBT	**7**	·	7
Comparison	**5**	·	5
Rectifiers	**32**	**10**	**42**
Three-phase	**14**	**5**	19
Full-wave	**8**	**5**	13
Half-wave	**10**	·	10
AC control and cycloconverters	**5**	·	**5**
Cycloconverters	**5**	·	5
DC-DC converters	**19**	**15**	**34**
Buck	**8**	**7.5**	15.5
Boost	**6**	**5**	11
SEPIC and Cuk	**5**	·	5
Field chopping	·	**2.5**	2.5
DC power supplies	**6**	·	**6**
Linear regulator	**6**	·	6
Inverters	**27**	·	**27**
PWM	**10**	·	10
Filters	**7**	·	7
Multilevel	**6**	·	6
Square-wave	**4**	·	4
Applications	**9**	·	**9**
Solar PV	**5**	·	5

Topic and subtopic	2020-21	Term tests	All
HVDC	**4**	·	4

3. Priority list:

Must know: Power devices (48), Rectifiers (42), DC-DC converters (34), Inverters (27), Power computations (19), Applications (9), DC power supplies (6), AC control and cycloconverters (5)

Likely: none

Less often: none

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. Power computations:

19 marks, 3 questions, in 1 of 1 papers. Book: Hart Ch. 2.

Most often numerical (2 of 3). Subtopics: Distortion; Nonlinear loads.

Formulas:

Subtopic	Formula	Equation	Used in
Distortion	Rms of a distorted wave	$I_{rms}^2 = \sum_n I_n^2$	20-21 1(a)
	Distortion factor	$DF = \dfrac{I_1}{I_{rms}}$	20-21 1(a)
	Total harmonic distortion	$THD = \dfrac{\sqrt{I_{rms}^2 - I_1^2}}{I_1}$	20-21 1(a)
	DF and THD	$DF = \sqrt{\dfrac{1}{1+THD^2}}$	20-21 1(a)
	Power factor	$pf = \dfrac{P}{S} = DF\cos\theta_1$	20-21 1(a)
	One harmonic only	$I_n = THD\, I_1$	20-21 1(d)
Nonlinear loads	Crest factor	$CF = \dfrac{I_{peak}}{I_{rms}}$	20-21 7(d)
	Power with a sine source	$P = V_1 I_1 \cos\theta_1$	20-21 7(d)

Questions:

No.	Paper	Marks	Type	Question
Distortion				
1	20-21 1(a)	7	derivation	Define distortion factor (DF) and total harmonic distortion (THD). Derive that $DF = \sqrt{\dfrac{1}{1+(THD)^2}}$

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No.	Paper	Marks	Type	Question
2	20-21 1(d)	6	numerical	A sinusoidal current of 10A rms at a 60 Hz fundamental frequency is contaminated with a ninth harmonic current. The current is expressed as $i(t) = 10\sqrt{2}\sin(2\pi 60t) + I_9\sqrt{2}\sin(18\pi 60t)$ A. Determine the value of the ninth harmonic rms current I9 if the THD is -i) 5 percent ii) 20 percent.
Nonlinear loads				
3	20-21 TT1 7(d)	6	numerical	A sinusoidal voltage source of $v(t) = 240\sqrt{2}\sin(2\pi 60t)$ V is applied to a nonlinear load, resulting in a nonsinusoidal current which is expressed in Fourier series form as $i(t) = 8\sin(2\pi 60t) + 4\sin(4\pi 60t)$ A. Determine (a) the power absorbed by the load, (b) the power factor of the load, (c) the distortion factor of the load current, (d) the total harmonic distortion of the load current, (e) crest factor of the load current.

4.2. Power devices:

48 marks, 8 questions, in 1 of 1 papers. Book: Rashid.

Most often theory (6 of 8). Subtopics: Diodes; Thyristors; IGBT; TRIAC and BJT; Comparison.

Formulas:

Subtopic	Formula	Equation	Used in
Diodes	Recovery time	$t_{rr} = t_a + t_b$	20-21 1(b)
	Peak reverse current	$I_{RR} = t_a \dfrac{di}{dt}$	20-21 1(b)
	Storage charge	$Q_{RR} \approx \dfrac{1}{2}\dfrac{di}{dt}t_{rr}^2$	20-21 1(c)
	Peak from the charge	$I_{RR} = \sqrt{2Q_{RR}\dfrac{di}{dt}}$	20-21 1(c)
Thyristors	**Two-transistor model**	$I_A = \dfrac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1-(\alpha_1+\alpha_2)}$	20-21 2(a)
	Turn-on condition	$\alpha_1 + \alpha_2 \approx 1$	20-21 2(a)

Questions:

No.	Paper	Marks	Type	Question
Diodes				
1	20-21 1(b)	6	theory	Define reverse recovery current and reverse recovery time for diode switching. What is the importance of these parameters?
2	20-21 1(c)	6	numerical	The manufacturer of a selected diode gives the rate of fall of the diode current di/dt=20 A/μs, and its reserve *[sic: should be reverse]* recovery time t_{rr}=5 μs. What value of peak reverse current and the storage charge do you expect?
3	20-21 TT1 7(a)	4	numerical	Explain the reverse recovery time of diode with diagram.
Thyristors				
4	20-21 2(a)	7	theory	Draw the two transistors model of a thyristor and explain the turn-on condition of a thyristor with equations from this model.
5	20-21 TT1 7(b)	5	theory	Draw the I-V curve of thyristor. Define Holding current and Latching current for thyristor
IGBT				
6	20-21 3(b)	7	theory	Draw the cross section of an IGBT showing the layers. Differentiate between punch-through and non-punch through IGBTs.

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No.	Paper	Marks	Type	Question
TRIAC and BJT				
7	20-21 6(c)	8	theory	Explain the structure of TRIAC with diagram. Mention the constructional feature of a Power BJT.
Comparison				
8	20-21 6(d)	5	theory	Compare the following device in terms of power capacity and switching speed: BJT, MOSFET, GTO, IGBT, MCT

4.3. Rectifiers:

42 marks, 6 questions, in 1 of 1 papers. Book: Hart Ch. 3, 4.

Most often numerical (5 of 6). Subtopics: Half-wave; Full-wave; Three-phase.

Formulas:

Subtopic	Formula	Equation	Used in
Half-wave	Impedance and angle	$Z = \sqrt{R^2 + (\omega L)^2},\ \theta = \tan^{-1}\dfrac{\omega L}{R}$	20-21 2(b)
	RL load current	$i = \dfrac{V_m}{Z}\sin(\omega t - \theta) + Ae^{-\omega t/\omega\tau}$	20-21 2(b)
	Constant	$A = -\dfrac{V_m}{Z}\sin(\alpha - \theta)e^{\alpha/\omega\tau}$	20-21 2(b)
	Average, half-wave	$I_o = \dfrac{1}{2\pi}\displaystyle\int_\alpha^\beta i\, d(\omega t)$	20-21 2(b)
	Rms, half-wave	$I_{rms}^2 = \dfrac{1}{2\pi}\displaystyle\int_\alpha^\beta i^2\, d(\omega t)$	20-21 2(b)
	Load power	$P = I_{rms}^2 R$	20-21 2(b)
	Half-wave output	$V_o = \dfrac{V_m}{2\pi}(\cos\alpha - \cos\beta)$	20-21 2(b)
Full-wave	**Controlled bridge, R load**	$V_o = \dfrac{V_m}{\pi}(1 + \cos\alpha)$	20-21 2(c)
	Rms, controlled bridge	$V_{rms} = \dfrac{V_m}{\sqrt{2}}\sqrt{1 - \dfrac{\alpha}{\pi} + \dfrac{\sin 2\alpha}{2\pi}}$	20-21 2(c)
	Diode bridge	$V_o = \dfrac{2V_m}{\pi}$	20-21 7(c)
Three-phase	**Diode bridge**	$V_o = \dfrac{3V_m}{\pi} = 0.955V_m$	20-21 3(a)
	Controlled bridge	$V_o = \dfrac{3V_m}{\pi}\cos\alpha$	20-21 3(d)
	Harmonics (controlled)	$V_n = \dfrac{3V_m}{\pi}\dfrac{2}{n^2 - 1}\sqrt{\cos^2\alpha + n^2\sin^2\alpha}$	20-21 3(d)
	Harmonics (diode)	$V_n = \dfrac{2V_o}{n^2 - 1}$	20-21 8(d)
	Harmonic current	$I_n = \dfrac{V_n}{\sqrt{R^2 + (n\omega L)^2}}$	20-21 3(d)

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Subtopic	Formula	Equation	Used in
	Rms load current	$I_{rms}^2 = I_o^2 + \sum_n \left(\frac{I_n}{\sqrt{2}}\right)^2$	20-21 8(d)
	Rms diode current	$I_{D,rms} = \frac{I_{rms}}{\sqrt{3}}$	20-21 8(d)
	Line current	$i_a = i_{D1} - i_{D4}$	20-21 3(a)

Questions:

No.	Paper	Marks	Type	Question
Half-wave				
1	20-21 2(b)	10	numerical	For the circuit of Fig. 2(b), the source is 120 V rms at 60 Hz, R=20 Ohm, L=0.04 H, and the delay angle is 45°. Determine (i) an expression for i(ωt) , (ii) the average current, (iii) the power absorbed by the load, and (iv) the power factor.
Full-wave				
2	20-21 2(c)	8	numerical	A full-wave controlled bridge rectifier has an ac input of 120 V rms at 60 Hz and a 20 Ω load resistor. The delay angle is 40°. Determine the average current of the load, the power absorbed by the load, the apparent power of the sourced *[sic: should be source]* and the input pf.
3	20-21 TT1 7(c)	5	numerical	With proper alignment, show the waveshape of output voltage, output current, diode currents and source current for a full wave bridge rectifier with sinusoidal source and R-L load.
Three-phase				
4	20-21 3(a)	8	theory	Draw the configuration of three phase uncontrolled rectifier. Show the source voltages, output voltage, diode currents and line currents with proper alignment.
5	20-21 3(d)	6	numerical	A three phase controlled rectifier has an input voltage of 480 V rms at 60 Hz. The load of the rectifier is resistor of 10 Ω in series with 50 mH inductance. (i) Determine the delay angle required to produce an average current of 50A in the load (ii) Determine the amplitude of the harmonics n=6 and n=12. For harmonics calculation, Fig. 3(d) can be utilized.
6	20-21 TT2 8(d)	5	numerical	A three phase rectifier is supplied by 480 V rms line to line 60 Hz source. The RL load is 100 Ω resistor in series with 15 mH inductor. Determine the rms load current and rms diode currents.

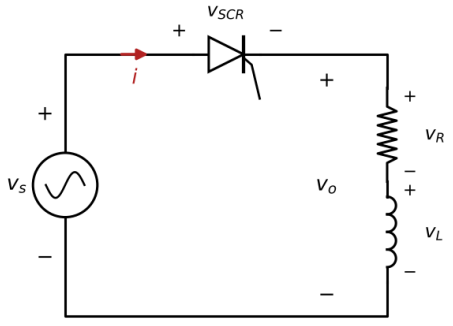


Figure 1: 20-21 2(b): Fig 2(b) of the paper (redrawn)

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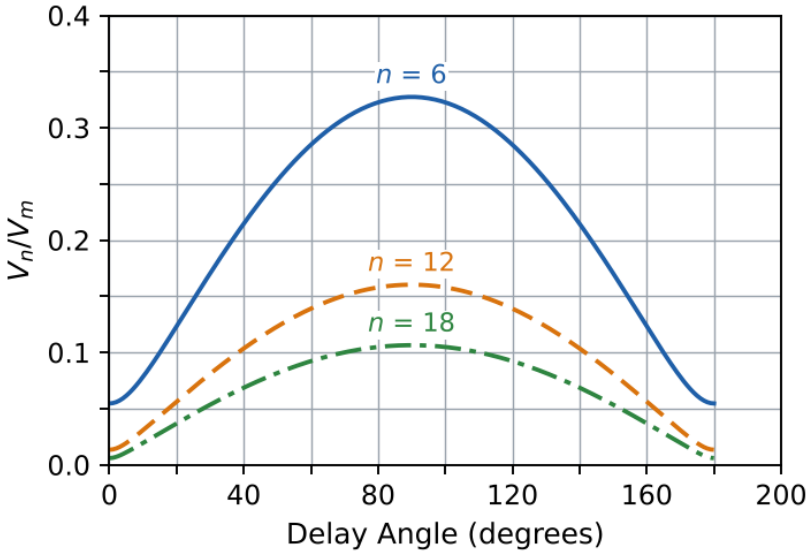


Figure 2: 20-21 3(d): Fig. 3(d) of the paper (redrawn)

4.4. AC control and cycloconverters:

5 marks, 1 question, in 1 of 1 papers. Book: Hart Ch. 5; Rashid.

Most often theory (1 of 1). Subtopics: Cycloconverters.

Questions:

No.	Paper	Marks	Type	Question
Cycloconverters				
1	20-21 6(b)	5	theory	What is cyclo-converter? Mention its applications.

4.5. DC-DC converters:

34 marks, 6 questions, in 1 of 1 papers. Book: Hart Ch. 6.

Most often theory (2 of 6). Subtopics: Buck; Boost; SEPIC and Cuk; Field chopping.

Formulas:

Subtopic	Formula	Equation	Used in
Buck	**Buck output**	$V_o = DV_s$	20-21 4(b)
	Buck ripple current	$\Delta i_L = \frac{V_o(1-D)}{Lf}$	20-21 4(b)
	Diode current	$I_D = (1-D)I_L$	20-21 4(b)
	Switch current	$I_S = DI_L$	20-21 4(b)
	Inductor extremes	$I_{max,min} = I_L \pm \frac{\Delta i_L}{2}$	20-21 8(b)
	Buck ripple voltage	$\frac{\Delta V_o}{V_o} = \frac{1-D}{8LCf^2}$	20-21 8(b)
Boost	**Boost output**	$V_o = \frac{V_s}{1-D}$	20-21 4(c)
	Boost inductor current	$I_L = \frac{V_o^2}{V_s R}$	20-21 4(c)

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Subtopic	Formula	Equation	Used in
	Boost ripple current	$\Delta i_L = \frac{V_s D}{Lf}$	20-21 4(c)
	Boost ripple voltage	$\frac{\Delta V_o}{V_o} = \frac{D}{RCf}$	20-21 4(c)
SEPIC and Cuk	SEPIC	$V_o = V_s \frac{D}{1-D}$	20-21 4(d)
	Cuk	$V_o = -V_s \frac{D}{1-D}$	20-21 4(d)
Field chopping	Field current	$I_f = \frac{\delta V_s}{R_f}$	20-21 8(c)
	Armature equation	$V_a = E_b + I_a R_a$	20-21 8(c)
	Speed	$\omega = \frac{V_a - I_a R_a}{K_v I_f}$	20-21 8(c)

Questions:

No.	Paper	Marks	Type	Question
Buck				
1	20-21 4(b)	8	derivation	Draw the circuit diagram and the waveform of inductor current, diode current and capacitor current for a typical buck converter. Also derive the relation between inductor current and diode current from the waveforms.
2	20-21 TT2 8(b)	7.5	numerical	A buck converter has parameters, Vs=50 V, D=0.4, R=20 Ω, L=400 μH and C=100 μF. The switching frequency is 20 kHz. a) Determine the output voltage, b) Determine the average, minimum and maximum inductor current c) The output ripple voltage.
Boost				
3	20-21 4(c)	6	design	A boost converter has an input of [unclear: 5 V] and an output of 25 W at 15 V. The minimum inductor current must be no less than 50% of the average. The ripple output must be less than 1 percent. The switching frequency is [unclear: 300 kHz]. (i) Determine the duty ratio. (ii) Minimum inductor value (iii) Minimum capacitor value
4	20-21 TT2 8(a)	5	diagram	Draw the circuit diagram, equivalent circuit for switch on and off condition and the waveforms of the inductor current, diode current and load voltage for a typical boost converter.
SEPIC and Cuk				
5	20-21 4(d)	5	theory	Differentiate between SEPIC converter and Ćuk converter.
Field chopping				
6	20-21 TT2 8(c)	2.5	theory	Draw the circuit diagram to control the speed of a separately excited dc motor by chopping field current supply. Explain the operation using required equations.

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4.6. DC power supplies:

6 marks, 1 question, in 1 of 1 papers. Book: Hart Ch. 7.

Most often theory (1 of 1). Subtopics: Linear regulator.

Formulas:

Subtopic	Formula	Equation	Used in
Linear regulator	Linear regulator	$V_o = V_{ref}\left(1 + \frac{R_1}{R_2}\right)$	20-21 4(a)
	Regulator loss	$P_{loss} = (V_s - V_o)I_L,\ \eta \approx \frac{V_o}{V_s}$	20-21 4(a)

Questions:

No.	Paper	Marks	Type	Question
Linear regulator				
1	20-21 4(a)	6	theory	Draw a typical circuit of a linear dc voltage regulator. What is the drawback of it? How can it be overcome?

4.7. Inverters:

27 marks, 4 questions, in 1 of 1 papers. Book: Hart Ch. 8.

Most often numerical (2 of 4). Subtopics: Multilevel; PWM; Square-wave; Filters.

Formulas:

Subtopic	Formula	Equation	Used in
Multilevel	Cascade output	$v_o = v_1 + v_2 + v_3$	20-21 5(b)
PWM	Modulation ratios	$m_a = \frac{V_{sine}}{V_{tri}},\ m_f = \frac{f_{carrier}}{f}$	20-21 5(c)
	PWM fundamental	$V_1 = m_a V_{dc}$	20-21 5(c)
	Harmonic current at the carrier	$I_n = \frac{V_n}{n\omega L}$	20-21 5(c)
Square-wave	Square-wave harmonics	$V_n = \frac{4V_{dc}}{n\pi}$	20-21 5(d)
	Square-wave THD	$THD = 48.3\ \%$	20-21 5(d)
Filters	Resonance	$\omega_0 = \frac{1}{\sqrt{LC}}$	20-21 6(a)
	Series filter reactance	$X_n = \omega_0 L\left(n - \frac{1}{n}\right)$	20-21 6(a)
	Load harmonic	$\frac{V_{o,n}}{V_1} = \frac{1}{n}\frac{R}{\sqrt{R^2 + X_n^2}}$	20-21 6(a)
	Capacitor peak	$V_{C,max} \approx \frac{I_1}{\omega_0 C}$	20-21 6(a)

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Questions:

No.	Paper	Marks	Type	Question
Multilevel				
1	20-21 5(b)	6	theory	Draw the circuit of a three-level cascade multi source inverter. Why pattern swapping is important in multi-source inverter circuit?
PWM				
2	20-21 5(c)	10	design	Design a bipolar PWM inverter that will produce a 75-V rms 60-Hz output from a 150 V dc source. The load is a series RL combination with R=12Ω and L=60 mH. Select the switching frequency such that the current THD is less than 10%. Assume the impedance at the carrier frequency is entirely inductive.
Square-wave				
3	20-21 5(d)	4	numerical	Determine the voltage THD of a square wave inverter.
Filters				
4	20-21 6(a)	7	numerical	A 10 Ohm resistive load requires a 1000-Hz, 50-V rms sinusoidal voltage. The THD of the load voltage must be no more than 5 percent. An adjustable dc source is available. (i) Design an inverter for this application. (ii) What is maximum voltage across the capacitor?

Normalized Fourier Coefficients V_n/V_{dc} for Bipolar PWM

	$m_a = 1$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n = 1$	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10
$n = m_f$	0.60	0.71	0.82	0.92	1.01	1.08	1.15	1.20	1.24	1.27
$n = m_f \pm 2$	0.32	0.27	0.22	0.17	0.13	0.09	0.06	0.03	0.02	0.00

Figure 3: 20-21 5(c): the table printed under Q5(c)

4.8. Applications:

9 marks, 2 questions, in 1 of 1 papers. Book: Rashid; Power Electronics Handbook.

Most often theory (2 of 2). Subtopics: HVDC; Solar PV.

Questions:

No.	Paper	Marks	Type	Question
HVDC				
1	20-21 3(c)	4	theory	Write down the advantages of DC power transmission.
Solar PV				
2	20-21 5(a)	5	theory	Briefly explain the types of switch mode DC-DC converters used in a stand-alone solar PV system.

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5. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Power computations				
1	20-21 1(a)	7	Distortion	Square the THD; add 1; use the definition of DF.
DC-DC converters				
2	20-21 4(b)	8	Buck	Read the diode current from the waveform; average it over a period; use the mean inductor current.

6. Repeat tracker:

No repeats among these papers.

7. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram	Design
Power computations	·	1	2	·	·
Power devices	6	·	2	·	·
Rectifiers	1	·	5	·	·
AC control and cycloconverters	1	·	·	·	·
DC-DC converters	2	1	1	1	1
DC power supplies	1	·	·	·	·
Inverters	1	·	2	·	1
Applications	2	·	·	·	·

8. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Power devices	48	25 %
2	Rectifiers	42	47 %
3	DC-DC converters	34	65 %
4	Inverters	27	79 %
5	Power computations	19	89 %
6	Applications	9	94 %
7	DC power supplies	6	97 %
8	AC control and cycloconverters	5	100 %

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